

Unit circle



1

a $\frac{1}{6}$

b $\frac{1}{2}$

c $\frac{5}{4}$

2

a Quadrant 4

b Quadrant 1

c Quadrant 2

d Quadrant 3

e Quadrant 1

f Quadrant 4

g Quadrant 3

h Quadrant 2

3 0° or 360° , 90° , 180° , 270°

4

a i Quadrant 1 and Quadrant 2

ii Quadrant 3 and Quadrant 4

b i Quadrant 1 and Quadrant 4

ii Quadrant 2 and Quadrant 3

c i Quadrant 1 and Quadrant 3

ii Quadrant 2 and Quadrant 4

5

a Quadrant 2

b Quadrant 2

c Quadrant 2

d Quadrant 1

6

a Positive

b Positive

c Negative

d Negative

e Negative

f Negative

g Positive

h Negative

i Negative

j Negative

k Positive

l Positive

m Positive

n Positive

o Negative

p Positive

7

a i $\frac{4}{5}$ ii $\frac{3}{5}$ iii $\frac{4}{3}$

b i $\frac{20}{29}$ ii $-\frac{21}{29}$ iii $-\frac{20}{21}$

c i $-\frac{21}{29}$ ii $\frac{20}{29}$ iii $-\frac{21}{20}$

8

a Yes

b Yes

c No

d Yes

9

a False

b False

c True

d True

e True



10

a True

b False

c True

d True

e False

11

a $2\sqrt{13}$

b

i $\frac{-3\sqrt{13}}{13}$

ii $\frac{-2\sqrt{13}}{13}$

iii $\frac{3}{2}$

12

a $\sin \theta = -\frac{4}{5}$

b $\tan \theta = -\frac{4}{3}$

13

a $\cos \theta = \pm \frac{3}{4}$

b $\tan \theta = \pm \frac{\sqrt{7}}{3}$

14

a $\cos \theta = -\frac{4}{5}$

b $\tan \theta = -\frac{3}{4}$

15

a $\cos \theta = \frac{10\sqrt{269}}{269}$

b $\sin \theta = -\frac{13\sqrt{269}}{269}$

16 $\cos \theta = -\frac{8}{17}$

17 a $\sin \theta = \frac{11}{61}$

b $\sin \theta = \frac{\sqrt{13}}{7}$

18 a $\tan \theta = \frac{2\sqrt{10}}{3}$

b $\tan \theta = \frac{1}{3}$

Equivalent ratios

19



20

a Quadrant 2

b Negative

c 30°

d $-\cos 30^\circ$

21

a Quadrant 4

b Negative

c 30°

d $-\tan 30^\circ$

22

a $\sin 87^\circ$

b $-\cos 15^\circ$

c $-\tan 61^\circ$

d $-\sin 60^\circ$

e $\sin 33^\circ$

f $-\tan 50^\circ$

g $\sin 40^\circ$

h $-\cos 65^\circ$

i $-\tan 15^\circ$

j $-\sin 55^\circ$

k $\cos 62^\circ$

l $-\tan 2^\circ$

m $-\sin 41^\circ$

n $-\cos 42^\circ$

o $\tan 71^\circ$

p $\sin 41^\circ$

q $-\cos 77^\circ$

r $-\tan 40^\circ$

23

a $-\frac{b}{a}$

b a

c b

d $-a$

e $-a$

f $-b$

24

a i $Q(-a, b)$

ii $R(-a, -b)$

iii $S(a, -b)$

b i $\sin 49^\circ$

ii $-\sin 49^\circ$

iii $-\sin 49^\circ$

c i $\sin x$

ii $-\sin x$

iii $-\sin x$

25

a i $Q(-a, b)$

ii $R(-a, -b)$

iii $S(a, -b)$

b i $-\cos 63^\circ$

ii $-\cos 63^\circ$

iii $\cos 63^\circ$

c i $-\cos x$

ii $-\cos x$

iii $\cos x$

26

a i $Q(-a, b)$

ii $R(-a, -b)$

iii $S(a, -b)$

b i $-\tan 62^\circ$

ii $\tan 62^\circ$

iii $-\tan 62^\circ$

c

i $-\tan x$

ii $\tan x$



iii $-\tan x$

27

a $\sin \theta$

b $\cos \theta$

c $\sin \theta$

d $-\cos \theta$

e $-\sin \theta$

f $\cos \theta$

g $-\sin \theta$

h $\cos \theta$

28

a 0.93

b 0.36

c -0.93

d 0.36

e -0.36

f -0.93

g 0.36

h -0.93

Use technology

29

a 0.56

b -0.59

c -0.34

d 0.49

e -0.16

f -0.79

g -0.68

h 57.29

i 0.33

j 0.33

k 0.57

l -0.33

30

a 1

b 1

c 1

d 1

31

a $(0.819, 0.574)$

b 73°

c $(-0.602, 0.799)$

32

a $(0.41, 0.91)$

b $(-0.41, 0.91)$

c 114°

d $(0.41, -0.91)$

e 294°